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Design and on-orbit performance evaluation of Ethiopian earth observation satellite multispectral optical imaging payload

Dinaol Gadisa^{a,*}, Eshet Tafes^{b,2}^a Department of Aerospace Engineering, Korean Advanced Institute of Science and Technology, KAIST, 291, Daejeon, 34141, Republic of Korea^b Department of Space Engineering, Space Science and Geospatial Institute, Gulelle, Addis Ababa, 33679, Ethiopia

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ABSTRACT

The Ethiopian Earth observation micro-satellite was successfully launched into a sun-synchronous orbit equipped with an optical multispectral imaging payload system. This optical multispectral imaging payload camera weighs 1.4 kg with four spectral bands: RGB and NIR, each having a unique spectral range. Furthermore, it has a field of view of $7.4^\circ \times 0.3^\circ$, a relative F-number of 9.8°, and a spectral range of 0.45–0.9 μm . The present article covers the design and on-orbit performance analysis of the ETRSS-1 electro-optical imaging payload system, which is based on the COTS compact and unobscured off-axis three-mirror anastigmatic system for high-resolution imaging. Additionally, selecting an appropriate imaging sensor, optomechanical architecture, and performance characteristics for optical instruments is investigated to evaluate spatial, spectral, and radiometric requirements and the payload's on-orbit performance and capability.

1. Introduction

Many remote sensing instrument utilized on Earth observation satellites has been developed, and many users have discovered that the data from this equipment is an invaluable source of information for Earth-related research and applications. Furthermore, the utilization of earth observation satellites equipped with Multispectral imaging payload has become increasingly important for monitoring climate and environmental changes, which is also the most common payload for remote sensing satellites [1].

The Ethiopian Earth observation satellite is the first to be launched into a sun-synchronous orbit for remote sensing and experimental applications. The satellite is equipped with a multispectral-optical imaging payload system designed to acquire high-resolution images for Earth observation and remote sensing applications such as water resources, desertification, vegetation, and other periodic observations, as well as the detection and assessment of natural disasters caused by climate change [2,3]. Fig. 1 represents the Ethiopian remote sensing satellite equipped with a multispectral optical imaging payload. This optical imaging payload has four multispectral channels and is capable of collecting images at a GSD of 13.75 m, an orbital altitude of 628.10 km, and a swath of 80 km.

Since the pioneering development of multispectral imaging cameras of the Landsat 1 (ERTS-1) earth observation mission, several studies

have contributed to the advancement of optical imaging payload design for a variety of mission applications, including earth observation and planetary body exploration.

Barnes et al. [4] reported the pre-launch features of the moderate resolution image spectroradiometer incorporated into the earth observation system (EOS)-AM1 spacecraft. The optical imaging system includes 36 bands with a geometric instantaneous field of view at nadir of 0.25, 0.5, and 1.0 km. The pre-launch testing results exhibit that the optical instrument's size, mass, power, and data rate, as well as its spatial performance and radiometric sensitivity, are all within the specified limits.

In addition, Wertz et al. [5] emphasized the overall process of designing and sizing optical payloads for Earth observation, whereas Cruise et al. [6] covered payload design challenges such as optics, electronics, thermal, structure, and mechanism.

Abolghasemi and Abbasi-Moghadam [1] presented the analysis and techniques for designing an experimental imaging payload for a remote sensing satellite. Furthermore, they investigated system performance by calculating the imaging payload's SNR detection and imaging MTF resolution at a spatial resolution of 50 m across three bands (green, red, and NIIR). The SNR and MTF of the payload push-broom scanner type flying at 700 km altitude exceeded 100 and 40, respectively, according to their findings.

* Corresponding author.

E-mail addresses: gdina@kaist.ac.kr, zdinaol2@gmail.com (D. Gadisa).¹ Ph.D. Candidate, Aerospace System and Control Lab.² Associate Researcher, Space engineering dept.

Nomenclature

Acronyms

CAST	China academy of space technology
CCD	Charged Couple Device
CMOS	Complementary metal-oxide semiconductor
DTS	Data transmission subsystem
EFL	Effective focal length
ESSTI	Ethiopian Space Science and Technology Institute
ETRSS-1	Ethiopian Remote Sensing Satellite-1
ERTS-1	Earth resource technology satellite (Landsat-1)
FOV	Field of view
GERD	Grand Ethiopian Renaissance Dam
GSD	Ground sampling distance
LEO	Low earth orbit
MSC	Multispectral Camera
MTF	Modulation transfer function
OBDH	On-board data handling
PM	Primary mirror
PSF	Point spread function
RMSE	Root mean square error
SM	Secondary mirror
SNR	Signal-to-noise ratio
TMA	Tri-mirror anastigmatic
TDI	Time delay integration

Symbols

a	Diameter of diffraction disk [mm]
A_d	Area of the detector [mm]
b	Number of bit quantization
c	Velocity of light in vacuum [m/s]
d	Pixel size of sensor [mm]
d_i	Thickness of the i th mirror [mm]
D	Aperture diameter of optical system [mm]
E_λ	Solar/lunar spectral radiance [W/m ²]
F_l	Effective focal length [mm]
G	Gravitational constant [N m ² /kg ²]
G_r	Ground resolution [mm]
h	Planck constant [kg m ² /s]
h_1, h_2, h_3	Heights of the marginal ray [mm]
L_p	Total scattered solar/atmosphere
$L(\lambda)$	Spectral radiance [W/sr m Hz]
k	Boltzmann constant [J/k]
M	Earth's mass [kg]
M_1, M_2, M_3	Primary, Secondary, Tertiary mirror
M_{cm}	MTF of the camera
M_o, M_e, M_c	Optical, detector, electronic MTF
q	Charge on electron [C]
Q	Quality factor
r_i	Curvature radius of the i th mirror [mm]
R	Spacecraft orbital altitude [km]
R_b	Data rate generated by sensor
R_0	Earth's radius [km]
S_t	Sampling time
S_e	Output signal
T	Absolute temperature [K]
T_{int}	Integration time
V_{sat}	Velocity of spacecraft [m/s]

Greek letter

β	Half-field angle [deg]
ϵ	Encoder noise bit degradation
η	Quantum efficiency of the detector
$\theta_1, \theta_2, \theta_3$	Paraxial angle of ray
θ_{max}	Maximum angular rate

θ_r	Angular diameter of resolution element
$\kappa_1, \kappa_2, \kappa_3$	Conic constant
ξ	On-board camera data storage capacity [Mbps]
λ	Operating spectral wavelength [nm]
σ	Noise standard deviation
σ_e, σ_q	Electronics and quantization noise
σ_p, σ_d	Photon, detector noise
ρ_λ	Spectral reflection coefficient of target
$\tau_0(\lambda)$	Transmittance of optical system
χ_1, χ_2, χ_3	Spherical aberration, Coma, Astigmatism

Jafarsalehi et al. [7] reported satellite imaging payload design optimization utilizing a genetic algorithm with aperture diameter and quality factor as design variables and constraints. Their findings included maximizing SNR and minimizing spherical aberration. Their analysis found that optimum aperture diameters were estimated for different sun elevation angles, and the optimal diameter reached ($D = 51$ cm) was 5 cm smaller than the conventional approach, with a decrease in SNR of 2.6 & 3.6 at 70 and 40 sun angles, respectively.

Recently, Alici et al. [8] presented a general-purpose unified framework for an end-to-end image chain simulation, enabling to analysis of the image quality of satellite imagers for earth observations, particularly during the design and test phases. They demonstrated the versatility of the generated image chain simulation through a variety of analyses, including the effects of sampling time and the amount of TDI elements.

Previous research efforts have mostly focused on a pre-launch optical imaging design performance evaluation or post-launch image data processing acquired by the optical imaging payload system. A robust top-down design and performance analysis approach has been proposed, starting with a definition of mission requirements and moving to the design and performance evaluation of a multi-spectral optical camera payload (i.e. pre-launch), as well as post-launch and a comprehensive investigation of image quality collected by the system.

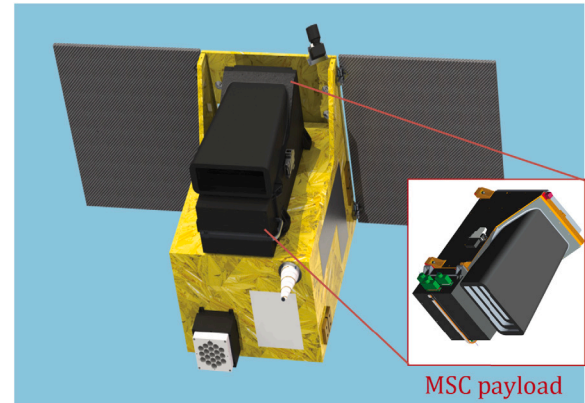


Fig. 1. Ethiopian remote sensing satellite with MSC payload (credit: ESSTI-CAST).

This paper discusses the design of the ETRSS-1 multi-spectral electro-optical imaging payload system, which is based on the off-axis TMA optical system, as well as pre- and post-launch performance evaluations. Additionally, the selection of a candidate imaging detector, optomechanical design within the micro-satellite form factor, and parameters that describe the capabilities and limitations of the optical instruments are examined to assess the system's performance, such as spatial resolution, modulation transfer function, noise characteristics, radiometric accuracy, and distortion levels.

The structure of this study is organized as follows: Section 2 describes the methodology in terms of multispectral optical imaging payload fundamental principles of optical system design insights and

requirement definitions, as well as the optomechanical system analysis, which includes both optical and structural design analysis. Section 3 describes the optical imaging payload camera of the Ethiopian Earth observation satellite's performance and onboard capability, which includes data collecting, radiation, and geometric calibration. Section 4 presents the conclusion to our work.

2. Methodology

2.1. Imaging payload design basics

For a given typical imaging payload, the radiation from a target image propagates through the atmosphere, and is mapped onto an image plane through collection optics, then sampled and converted into electronic signals for transmission, data processing and display. However, the actual configuration and implementation of an optical imaging payload chain are determined by the top-level system requirements [1]. Fig. 2 represents the top-down design and performance evaluation of the Ethiopian earth observation satellite's optical multispectral imaging payload, which will be explored further.

In addition, an optical imaging payload application often comprises a set of well-defined criteria for which they will be utilized, and these can be transformed into specific engineering requirements, or specifications, which drive the payload design. The essential requirements that a optical imaging payload must meet can be classified into three categories [9]: spatial resolution and field of view, light (radiance) sensitivity and brightness (or emittance) range, and spectral sensitivity and coverage.

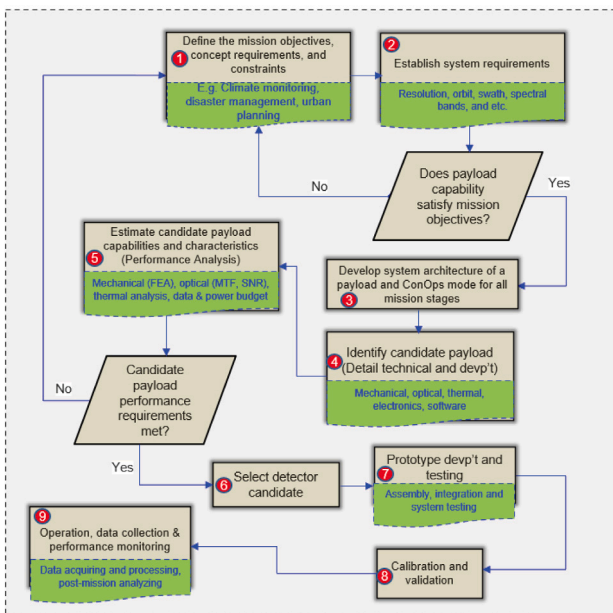


Fig. 2. Block diagram of ETRSS-1 optical MSC imaging payload design and performance evaluation.

Spatial requirements are among the simplest to specify, and depending on the mission applications, they must contain the minimum and maximum range to objects of interest, sensor field of view, and minimum size resolution in the objects of interest. The range between the objects of interest in the scene may vary, but in general, the maximum range is the limiting element in defining necessary sensor resolution, while the minimum range restricts the field of view.

Another important optical design requirement is the sensor's color sensitivity, as the quantity of optical energy available from a scene is determined by the width of the spectral wavelength area that the sensor employs to record images. This energy variation is caused by a variety of reasons, including spectral variations in solar radiation, detector sensitivity, and spectrum variations in spectral filter transmittance.

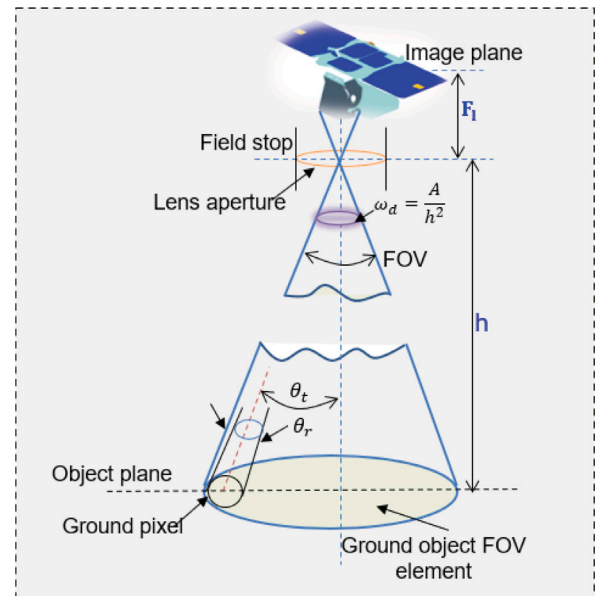


Fig. 3. An optical MSC imaging payload characteristics of the ETRSS-1 spacecraft.

Typically, color sensitivity is determined mostly by the application and has significant effects on optical payload design.

Radiometric requirements also affect the performance of an optical system used for investigating an application. These requirements include brightness sensitivity, brightness range, and calibrated sensor accuracy, which are significantly large as a manufacturing and testing constraints, but the first two are what drive the initial design or selection of an existing sensor for an application. Ultimately, the overall mission requirements determines the technical performance of the optical payload, and establishing performance thresholds provides a framework for balancing performance across multiple design elements, as technical specifications and payload operational concepts are frequently interconnected. Optical imaging payload performance evaluation categories include physical performance constraints such as spatial, spectral, radiometric, and temporal resolution, and operational constraints such as sensor duty cycle limits, tasking and scheduling limits [5].

To develop an optical imaging payload that meets a specific set of requirements, it is important to understand the key design factors as well as available technologies. The conceptual optical geometric characteristics of the Ethiopian Earth observation satellite depicted in Fig. 3 and the relationship between various variables determine the spacecraft's mission and objective, as well as the performance of its optical imaging payload.

There are various ways to define an optical imaging system. Furthermore, the performance of an optical imaging payload system was determined by a variety of optical characteristics, including aperture size, field of view, optical aberrations, modulation transfer function, and others, all of which influence the quality and accuracy of the acquired images. Another important characteristic that influences the performance of optical imaging systems is diffraction, which is an essential limitation to the resolution of an optical system defined in angular terms. Regardless of the lens or mirror's quality, diffraction, or light bending, occurs at the optical system's edge. It causes the image of a point source of light to appear as a series of concentric rings that gradually fade away from the center, also known as the diffraction disk or point spread function.

2.1.1. Aperture diameter and resolution

The aperture is the hole in an optical imaging camera that allows light to pass through the film, and the amount of light that passes

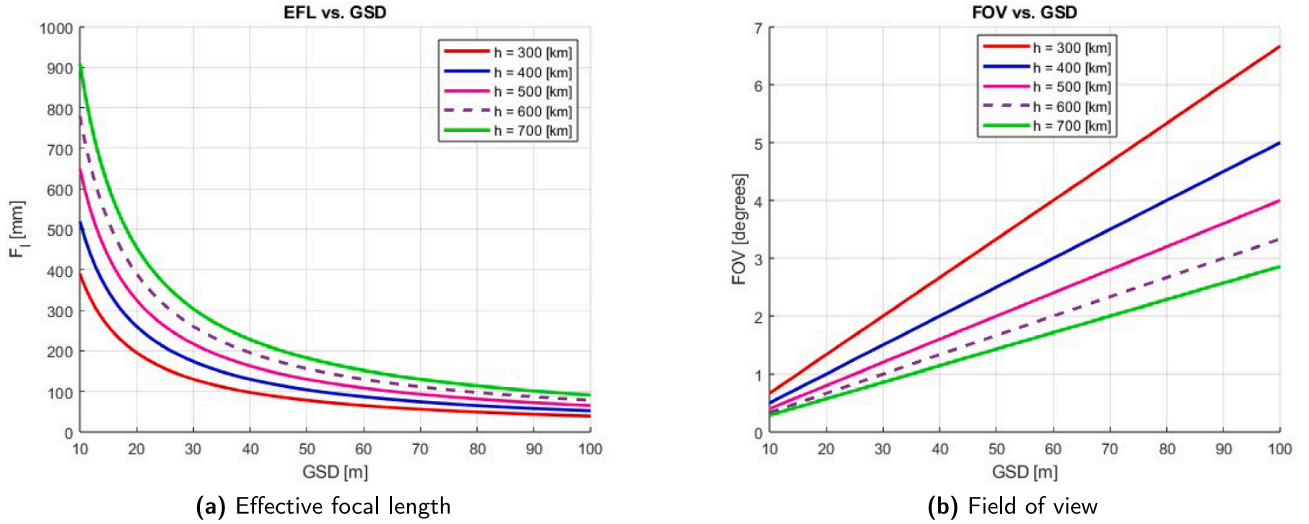


Fig. 4. GSD characteristics at different altitudes (for ETRSS-1 MSC pixel size, $d = 13 \mu\text{m}$).

through the aperture determines how the image appears. The larger the aperture, the more light can be gathered, resulting in higher-quality images [7].

The angular distance, θ_r , from the maximum at the center of the image to the first dark interference ring, also known as the Rayleigh limit or Rayleigh diffraction criteria, is reported in Ref. [10]:

$$\theta_r = 1.22 \frac{\lambda}{D} \quad (1)$$

where λ is the operating wavelength and D is the aperture diameter of the optical instrument. For the majority of operating systems, the optical imaging system can be considered near diffraction limited and in focus, leading in a diffraction disk or airy disk.

For a satellite at altitude, R , the linear resolution or ground resolution, G_r (diameter on the ground corresponding to point spread function, a , on the focus plane) at nadir is $G_r = 2.44 \left(\frac{R\lambda}{D} \right)$. The ground resolution (G_r) of an earth observation satellite's optical imaging payload system is often correlated with the diffraction-limited resolution of an optical imaging payload, which is the lowest discernible detail on the Earth's surface that the system can capture. It is determined by the system's optical design, which includes focal length, aperture size, detector pixel size, and satellite altitude. Further, it is proportional to the spacecraft's orbital altitude and spectral wavelength, and inversely proportional to the system's aperture diameter.

An additional design parameter, the quality factor, Q , had been utilized while developing an optical imaging payload system with a detector array. It is defined as the ratio of pixel size (d) to the diameter of the diffraction disk or point spread function (a), (i.e. $Q = \frac{d}{a}$), which typically ranges from 0.5 to 2. For $Q < 1$, pixels are smaller than the diffraction disk and resolution is limited by diffraction optics, resulting in the best image resolution for a given aperture, but for $Q > 1$, resolution is limited by pixel size [7]. The pixel size, d , can be derived as reported in Ref. [5].

$$d = 2.44 \left(\frac{\lambda F_l}{D} \right) Q \quad (2)$$

where F_l is effective focal length.

2.1.2. Field of view and focal length

Field of view is another important performance characteristic of optical imaging that describes the angular field in the object space obtained by electro-optical systems. It is determined by the wavelength of interest and the angular resolution of sensor optics. In remote sensing

satellites, for a normal optical sensor focused at infinity, the field of view can be calculated as:

$$\text{FOV} = 2 \times \tan^{-1} \left(\frac{\text{sensor size}}{2F_l} \right) \quad (3)$$

where the focal length, F_l , determines the length of the optical collection system, and it relates to the radius of curvature of the lens surface. For a single lens, it is equivalent to the lenses surface's radius of curvature [5]; for a spherical reflecting surface, it is one-half of its radius; and for a parabolic surface, the quantity F_l equals the distance from the focus to the nearest point on the reflecting surface.

Additionally, an optical imaging system can be determined by its so-called infinity F-number or F-stop, which relates to the optical system focal length, as $F/\# = \frac{F_l}{D}$, where D is the aperture or diameter of the lens. The $F/\#$ is also used in determining the cone angle of all rays that the detector accepts.

Of course, there is a strong correlation between ground resolution, satellite velocity, and an optical imaging sensor's sampling time. As a result, the scanning sensor's sample time varies with satellite orbit. It is important to emphasize that Ethiopian earth observation satellite is in the circular low-orbit altitude and velocity of this spacecraft (V_{sat}), determined as described in Ref. [1].

$$v_{sat} = \sqrt{\frac{GM}{r_0 + h}} \quad (4)$$

where G is the gravitational constant. At the closest approach point, the spacecraft moves perpendicular to the ground station's line of sight. subsequently, the maximum angular rate of the satellite as viewed from the ground station, $\dot{\theta}_{max}$, can be computed as:

$$\dot{\theta}_{max} = v_{sat} \frac{r_0}{r_0 + R} \quad (5)$$

However, the spacecraft's altitude, the pixel size of the detector, and the focal length of the optical payload will help in calculating the Ground Sampling Distance (GSD) at the nadir, which is calculated as:

$$GSD = \frac{d \times R}{F_l} \quad (6)$$

where d is the pixel size of the detector, CCD.

Fig. 4 represents the effective focal length and the field of view for the Ground Sampling Distances at different possible spacecraft altitudes and pixel size of $d = 13 \mu\text{m}$. Table 1 shows that at a fixed EFL (F_l) of the ETRSS-1 MSC optical imaging camera payload with an effective focal length of 608 mm, lower altitudes provide higher ground sampling distance. This implies that in Fig. 4 the parameter GSD is contingent on the effective focal length and altitude of the spacecraft. The swath

Table 1

GSD of the ETRSS-1 MSC optical camera with fixed effective focal length and field of view.

CCD type	No. CCD [pixel/band]	f [mm]	FOV [°]	h [km]	GSD [m]	W [km]
AT71547 (E2V)	600	608	7.317	500	10.66	63.93
				645	13.75	82.47
				700	14.92	89.50

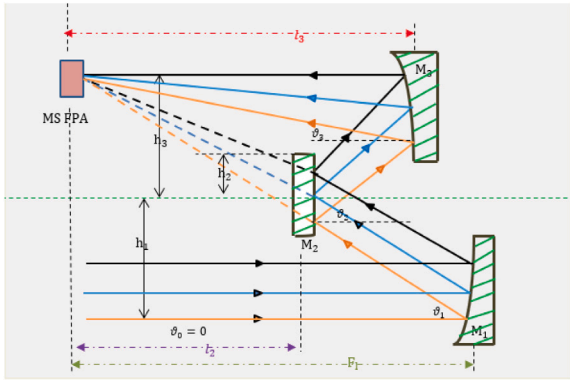
width of an optical imaging payload on an Earth observation satellite is another important element that determines the size of the ground coverage area in one image pass and the revisit time of the satellite. It is determined by the satellite's altitude, optical system field of view, and Earth's surface geometry, where the ideal sampling time is computed by considering the ground sampling distance and satellite ground velocity, described in Ref. [8]:

$$S_t = \frac{GSD}{\dot{\theta}_{max}} = \left(\frac{d \cdot R}{F_l} \right) \left(\frac{r_0 + R}{v_{sat} \cdot r_0} \right) \quad (7)$$

2.1.3. Aberration, coma, and astigmatism

The off-axis TMA optical imaging system offers various advantages as a payload system in Earth observation missions for multiple applications due to its unique characteristics of compactness, no central obstruction, a high degree of freedom, loose heat tolerance, and free of chromatic aberrations [11]. These characteristics have potential to provide a good optical solution to meet the requirement of high resolution while maintaining a wide field of view [12,13].

The widely used off-axis TMA configuration typically offsets the aperture stop or biases the FOV to avoid obstruction and achieve a wide FOV, making non-rotationally symmetric aberrations dominated by astigmatism and coma caused by the off-axis structure difficult to eliminate and posing manufacturing and alignment challenges [14]. However, the advancement of aberration theory, optical system optimization technique, manufacturing technology, and use of freeform surfaces have shown significant potential in minimizing asymmetric aberration to a certain extent caused by an off-axis optical system [15].

**Fig. 5.** Schematic diagram of the three-mirror system.

Assuming the object plane is at infinity and the stop is at the primary mirror, where h_1 , h_2 , and h_3 and θ_0 , θ_1 , θ_2 , and θ_3 are the heights of the incident marginal ray and paraxial angles of the ray after reflection (see Fig. 5), respectively. The obscure ratios of m_1 and m_2 , a_1 , a_2 , and the curvature of i th different mirror radii r_i and its related thickness (d_i) and transverse magnifications (m_i) are analyzed using paraxial optical theory given as:

$$\begin{bmatrix} a_1 & m_2 \\ a_2 & m_3 \\ r_i & d_i \end{bmatrix} = \begin{bmatrix} \frac{h_2}{h_1} & -\frac{\theta_1}{\theta_2} \\ \frac{h_3}{h_2} & -\frac{\theta_2}{\theta_3} \\ -\frac{2h_i}{\theta_i + \theta_{i-1}} & \frac{h_{i+1} - h_i}{\theta_i} \end{bmatrix} \quad (8)$$

It is important to note that the specifications of the optical imaging payload design always include the effective focal length (F_l), f-number ($F/\#$), and half field angle (β). Thus, some ray parameters can be easily determined from these specification, such as:

$$\begin{bmatrix} h_1 & \theta_1 \\ h_3 & \theta_3 \end{bmatrix} = \begin{bmatrix} \frac{F_l}{F/\#} & -\left(\frac{h_2 + h_1}{d_1}\right) \\ h_2 + d_2\theta_2 & \pm \frac{2}{F/\#} \end{bmatrix} \quad (9)$$

The third order of spherical aberration (χ_1), coma (χ_2), and astigmatism (χ_3) of this optical mirror system are linear functions of conic constants (κ_1 , κ_2 , and κ_3), which can be computed as presented in Ref. [14,15]:

$$\begin{bmatrix} \chi_1 \\ \chi_2 \\ \chi_3 \end{bmatrix} = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} + \begin{bmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{bmatrix} \begin{bmatrix} \kappa_1 \\ \kappa_2 \\ \kappa_3 \end{bmatrix} \quad (10)$$

where the following equations represent the variable z_1 , z_2 , and z_3 (Eq. (11)), u_1 , u_2 , and u_3 (Eq. (12)), v_1 , v_2 , and v_3 (Eq. (13)), as well as w_1 , w_2 , and w_3 (Eq. (14)), respectively.

$$\begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = - \begin{bmatrix} \frac{1}{4} h_1 \theta_3^3 (-m_3^3 S_{1f} + a_1 a_2 S_{1r}) \\ \frac{1}{4} H \theta_3^2 (m_3^2 S_{2f} + \mathcal{N} S_{1r}) \\ \frac{H^2}{4 h_1} (-m_3 S_{3f} + \frac{S_{3r} + 2\mathcal{N} S_{2r} + \mathcal{N}^2 S_{1r}}{a_1 a_2}) \end{bmatrix} \quad (11)$$

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = - \begin{bmatrix} \frac{1}{4} h_1 \theta_3^3 (-m_3^3 m_3^3) \\ 0 \\ 0 \end{bmatrix} \quad (12)$$

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = - \begin{bmatrix} \frac{1}{4} h_1 \theta_3^3 (a_1 (m_2 - 1)^3 m_3^3) \\ \frac{1}{4} H \theta_3^2 \left(\frac{(a_1 - 1)(m_2 - 1)^3 m_3^2}{m_2} \right) \\ \frac{H^2}{4 h_1} \left(\frac{(a_1 - 1)^2 (m_2 - 1)^3 m_3}{a_1 m_2^2} \right) \end{bmatrix} \quad (13)$$

$$\begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = - \begin{bmatrix} \frac{1}{4} h_1 \theta_3^3 (a_1 a_2 (m_3 - 1)^3) \\ \frac{1}{4} H \theta_3^2 (\mathcal{N} (m_3 - 1)^3) \\ \frac{H^2}{4 h_1} \left(\frac{\mathcal{N}^2 (m_3 - 1)^3}{a_1 a_2} \right) \end{bmatrix} \quad (14)$$

and the conic constant coefficients correcting the third order spherical aberrations, χ_1 , coma, χ_2 , and astigmatism, χ_3 are given:

$$\begin{bmatrix} \kappa_1 \\ \kappa_2 \\ \kappa_3 \end{bmatrix} = \begin{bmatrix} -\frac{m_3^3 S_{1f} + a_1 a_2 S_{1r} + a_1 (m_2 - 1)^3 m_3^3 \kappa_2 + a_1 a_2 (m_3 - 1)^3 \kappa_3}{m_2^3 m_3^3} \\ -\frac{m_2^2 (a_1 a_2 m_3 S_{3f} + \mathcal{N} (m_2^2 S_{2f} - S_{2r}))}{(a_1 - 1)(m_2 - 1)^3 m_3 (\mathcal{N} m_2 m_3 - a_1 (a_1 - 1))} \\ -\frac{m_2 (m_2^2 S_{2f} + S_{2r} + \mathcal{N} S_{1r}) + (a_1 - 1)(m_2 - 1)^3 m_3^3 \kappa_2}{\mathcal{N} m_2 (m_3 - 1)^3} \end{bmatrix} \quad (15)$$

where Eq. (16), (17), and (18) represent the parameters S_{1f} , S_{1r} , and $\mathcal{N}$ respectively, along with $i = 1, 2, 3$.

$$\begin{bmatrix} S_{1f} \\ S_{2f} \\ S_{3f} \end{bmatrix} = \begin{bmatrix} m_3^2 + a_1 (1 + m_2)^2 (1 - m_2) \\ -2m_2^2 - \frac{(1 - m_2^2)(m_2 - 1 + (1 + m_2)a_1)}{m} \\ 4m_2 + \frac{(1 - m_2)(m_2 - 1 + (1 + m_2)a_1)^2}{m_2^2 a_1} \end{bmatrix} \quad (16)$$

$$\begin{bmatrix} S_{1r} \\ S_{2r} \\ S_{3r} \end{bmatrix} = \begin{bmatrix} -(1 + m_3)^2 (1 - m_3) \\ -2(1 - m_3^2) \\ -4(1 - m_3) \end{bmatrix} \quad (17)$$

$$\mathcal{N} = \frac{a_2 (a_1 - 1 + m_2) - m_2}{m_2 m_3^3} \quad (18)$$

To summarize, the Ethiopian earth observation satellite multispectral optical payload camera system used the off-axis TMA optical system to correct three types of geometric aberrations: spherical aberration, coma, and astigmatism in the absence of specific symmetry conditions [16].

2.1.4. Data throughput

Data rate: The data rate required for Earth observation payloads is contingent upon the resolution, coverage area, amplitude accuracy, and several sensors [17]. When the spacecraft is in low-Earth orbit, its motion allows for easy scanning of the Earth perpendicular to the orbit plane. Furthermore, the sensor provides an image made up of the minimum resolvable pieces known as pixels, and the data rate generated by the sensor is calculated as reported in Ref. [18]:

$$R_b = \frac{n \times n_b \times N}{t} \quad (19)$$

where n is the number of detectors, n_b digital quantization, N number of bands, and t the integration time.

For 6000 detectors, four bands, and 10 digital quantizations with a minimal integration time, the data rate of the Ethiopian Earth observation satellite payload (auxiliary included) is around 134 Mbps, with a download rate of 100 Mbps.

Data storage: As the current earth observation industry and demand for high-quality imaging data grows, spatial and spectral resolution rise, meaning more data and pixels are processed, while the spacecraft's get smaller resulting in higher complexity (and less power to transmit data) [19].

It is apparent that downloading the appropriate amount of imagery acquired via an optical imaging payload while maintaining high throughput requires a significant amount of time and money. In addition, the storage capability is not useful if the data rates are not increased accordingly. It is trivial to show that there is a linear relationship between storage capability and data rate of optical imaging camera if the constraint to store all the images that can be downloaded in one access to ground station [20]:

$$\xi = \frac{15}{2} \left(\frac{\min}{\text{access}} \right) R_b \quad (\text{Mbps}) \quad (20)$$

Thus, the real restricting factor is data rates, not data storage, and so reduced data latency, intelligent on-board processing, and downloading selected and required regions of interest and translating it into insight and intelligence must be accounted for [19]. The MSC of the Ethiopian Earth observation satellite has a data storage capacity of more than 512 GB.

2.2. MSC optical camera specs

Different remote sensing applications requires various techniques for cost-effective missions [21]. The Ethiopian Earth observation satellite mission was intended to enhance Ethiopia's remote sensing application capabilities in climatic variability by providing periodicity observations in a range of fields. Following this mission requirements, the spacecraft has been developed based on the standardized satellite platform of CAST [2], which was equipped with an optical multi-spectral camera and data transmission subsystem. Furthermore, an optical imaging payload system's primary role is to collect, electro-optical convert, electronic process and multiplexing, digital data transmission, image reconstruction, and display electromagnetic radiation-generated object plane image information [10].

The Ethiopian Earth observation satellite's multi-spectral camera module collects images in nadir or roll maneuvers less than $\pm 40^\circ$ and at spacecraft attitude yaws 90° . The imaging data is subsequently transferred to the X-band ground station via the data transmission subsystem [2]. However, the MSC optical imaging payload of this spacecraft has been constrained by various variables such as orbit, attitude (e.g., during imaging mode and yaw $\pm 90^\circ$, and global attitude), interface, and alignment. The spacecraft operates from a sun-synchronous orbit at an altitude of 628.10 km, with a descent nodal local time of 10:30 AM, a revising period of 4 days, and a ground sampling distance of more than 13.75 m [2,3].

Additionally, the spacecraft's MSC payload is intended to acquire imagery data within a cumulative time duration of camera power of no

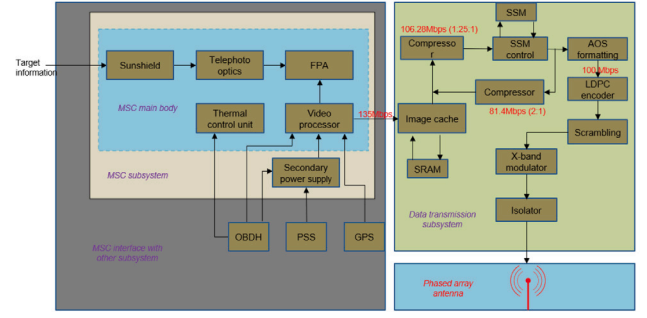


Fig. 6. Schematic of ETRSS-1 MSC composition & interface.

more than 8 min in one orbit, and no more than 70 min in one day. During the imaging mode, the ETRSS-1 spacecraft attitude determination accuracy was $\leq 0.05^\circ$, (3σ), pointing accuracy of $\leq 0.0^\circ$, (3σ), pointing stability of $\leq 0.005^\circ/\text{s}$, (3σ), attitude maneuver of $\pm 35^\circ$, while yaw $\pm 90^\circ$ mode, it was $\leq 0.1^\circ$, (3σ), $\leq 0.2^\circ$, (3σ), $\leq 0.01^\circ/\text{s}$, (3σ). While, in the global attitude acquisition mode, the spacecraft axis-Z was pointing to the sun, which should be considered in the MSC optical payload thermal interface design of mounting surface main body (-15°C ~ 30°C), electronics (-15°C ~ 50°C), and outside of payload cabin (-90°C ~ 90°C).

The interface requirements of the MSC optical camera of ETRSS-1 spacecraft encompass a set of specifications and standards necessary for seamless integration, operation, and compatibility with various subsystems and ground infrastructure. The flatness of the MSC optical camera of the ETRSS-1 spacecraft installation plane was ≤ 0.03 mm having a power supply of $+23\sim 29$ V with onboard data handling CAN baud rate of 500 kbs and current absorbability of ≥ 200 mA. Fig. 6 depicts the detailed composition and information or data flow the ETRSS-1 MSC optical imaging camera module and its interface with other subsystem.

The ETRSS-1 MSC optical imaging payload will receive broadcast data from the onboard data handling system, such as satellite time, attitude, star sensor, and GPS location. Moreover, the GPS receiver will supply one pulse per second for the MSC system and broadcast the time (second) corresponding to the pulse every second via CAN in 50 ms.

2.3. Optomechanical design and analysis

2.3.1. MSC optical imaging design

The most widely used earth observation missions rely substantially on electro-optical imaging payloads that operate in the visible and near-infrared spectral range. The actual configuration and implementation of a given optical imaging design are strongly contingent on the top-level requirements and resulting top-down system specification [7].

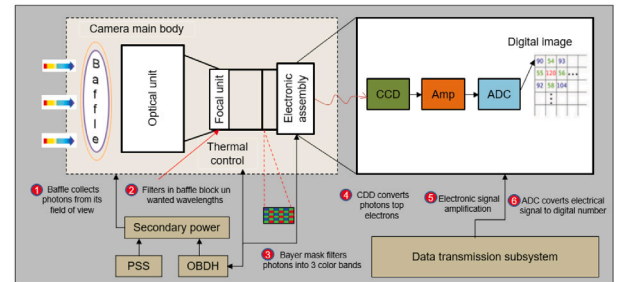


Fig. 7. Schematic chain of ETRSS-1 S/C MSC optical system and camera.

Fig. 7 represents the major functions associated with the Ethiopian earth observation satellite MSC optical imaging payload chain. The system is comprised of up of the camera's main body, that includes the optical unit, electronic assembly, and other units that communicate with the power supply system, onboard data handling system, and data

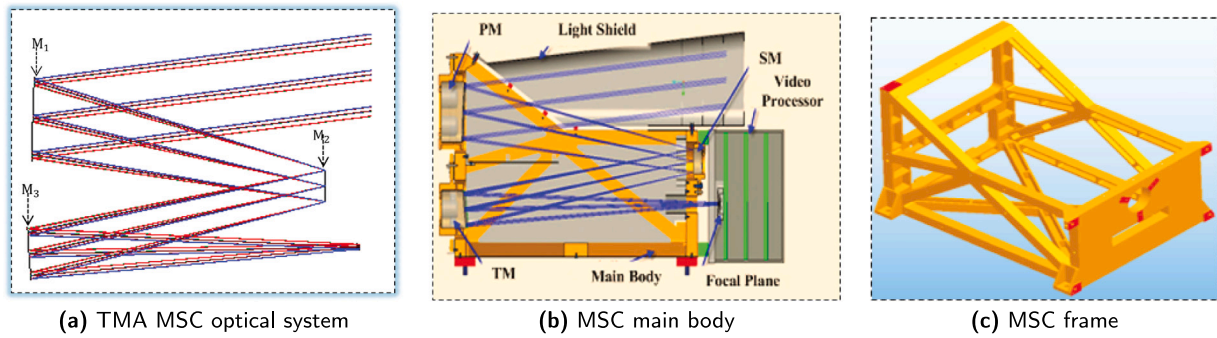


Fig. 8. ETRSS-1 MSC optical imaging payload components (credit: ESSTI-CAST).

Table 2

Summary of the ETRSS-1 MSC optical camera parameters.

Item name	Parametric value
Mass	≤ 15.5 kg
Focal length	608 mm
Spectrum	Blue (B1): $0.45 \mu\text{m} \pm 0.02 \mu\text{m} \sim 0.52 \mu\text{m} \pm 0.02 \mu\text{m}$ Green (B2): $0.52 \mu\text{m} \pm 0.02 \mu\text{m} \sim 0.59 \mu\text{m} \pm 0.02 \mu\text{m}$ Red (B3): $0.63 \mu\text{m} \pm 0.02 \mu\text{m} \sim 0.69 \mu\text{m} \pm 0.02 \mu\text{m}$ NIR (B4): $0.77 \mu\text{m} \pm 0.02 \mu\text{m} \sim 0.89 \mu\text{m} \pm 0.02 \mu\text{m}$
Detector	$6000 \times 13 \mu\text{m}$
Field of view	≥ 7.317 degree
Relative distortion	$\leq 1.5\%$
Static MTF	≥ 0.20 at 38.5 lp/mm
Dimension	$\leq 520 \times 290 \times 285$ mm
SNR	≥ 26.7 ($\alpha = 20^\circ$, $\rho = 0.05^a$) ≥ 48.5 ($\alpha = 70^\circ$, $\rho = 0.65$)
Power consumption	≤ 25 W (does not include thermal power consumption)
Data quantization	10 bits
Working time limit	≤ 10 min/orbit
Auxiliary telemetry data downlink	Image telemetry data down-linked in X-band GPS raw, Star sensor, & Gyro data

^a α : Solar zenith angle and ρ : Surface albedo.

transmission subsystem. The electromagnetic radiation from a target image propagates through the atmosphere and is mapped onto an image plane through collection optics, then sampled, and converted into electronic signals for transmission, data processing, and display [1].

The Ethiopian Earth observation satellite comprises the COTS off-axis three-mirror anastigmatic optical system with a reflective optical size of $510 \times 285 \times 290$ mm, a field of view of $7.4^\circ \times 0.3^\circ$, and an F-number of 9.8 [2], as well as a frame CCD camera with a Bayer mask filter. The CCD converts filtered photons into electronics, which are amplified to produce a digital number for each pixel in each band, and then interfaces with the data transmission and other subsystems. Moreover, as illustrated in Fig. 7 of the optical imaging system of the Ethiopian earth observation satellite, light from an object enters an off-axis tri-mirror astigmatism optical imaging system, where it is reflected and focused on a secondary mirror to avoid central obstructions. The secondary mirror, which is also concave, directs light to a tertiary mirror, which corrects any remaining optical imperfections like astigmatism. This tertiary mirror then reflects light into the focal plane, where the detector array is positioned. This configuration allows for a wide field of view and high-resolution imaging by preventing optical aberrations and ensuring that light channels remain unobstructed, resulting in clean, sharp images of the Earth's surface. Table 2 represents the optical budget of the Ethiopian Earth observation satellite MSC optical imaging payload, which includes optical dimensions as well as system weight.

2.3.2. MSC structural design

The Ethiopian Earth observation MSC optical imaging payload structure were adjusted to new designs or components while avoiding

Table 3

ETRSS-1 MSC optical imaging payload thermal design.

Components	Requirements	Parts	Thermal design	
			High temp. [$^\circ\text{C}$]	Low temp. [$^\circ\text{C}$]
Optical lens	20 ± 3 $^\circ\text{C}$	PM	19.4~19.6	19.1~19.2
		SM	19.4~19.8	19.2~19.5
		TM	19.4~19.6	19.1~19.3
Main body	20 ± 3 $^\circ\text{C}$	Frame	18.5~21.2	18.5~21.2
Detector	$0 \sim 25$ $^\circ\text{C}$	CCD	18.5~20.5	18.5~20.1

additional constraints on other components. Several essential tasks, ranging from requirement specification to optomechanical design and analysis, are required to ensure the system functionality, stability, as well as efficiency of this spacecraft's MSC imaging payload.

The ETRSS-1's MSC optical imaging payload structure is designed to support and protect optical components, maintain structural integrity, and maintain mechanical stresses encountered during launch and on-orbit operation, as shown in Fig. 8b and c.

Figs. 8 and 9 depict the structural components, stress analysis (Fig. 9a) of about 199 Hz, and information regarding the thermal characteristics of the MSC of Ethiopia's Earth observation satellite. The thermal requirements demand that the optical imaging payload components be kept within the acceptable temperature ranges specified in Table 3 during all operation phases. In addition, the ETRSS-1 MSC optical imaging payload main body is mounted horizontally, consisting of the main structural frame, primary mirror, secondary mirror, tertiary mirror, MS FPA, and video processor. The FPA and video processor are inside the main body of the MSC optical imaging payload and connected with other electronics through the cables. Its functions include receiving and distributing PPS; receiving, analyzing, transmitting, and executing the bus command from the satellite through the CAN bus; inspecting and gathering the camera's telemetry capacity and sending it to the satellite through the bus after coding; completing the analog signal processing of image data, adding the auxiliary data and sending to data transmission subsystem.

The secondary power supply is outside of the ETRSS-1 MSC optical imaging main body, installed in the electronic cabin of the spacecraft. Its functions include changing and generating the secondary power supply; receiving direct commands from the spacecraft (23~29 V) and transferring the power to the focal plane and video processor component. The external size of the secondary power supply of the ETRSS-1 MSC optical imaging payload is about $180 \times 180 \times 50$ mm and weighs about 2 kg.

2.4. Imaging sensor selection

The spacecraft in LEO moves with respect to the scene, and its optical instrument point moves at approximately 7 km/s across the Earth's surface. This motion can replace one of the scan dimensions,

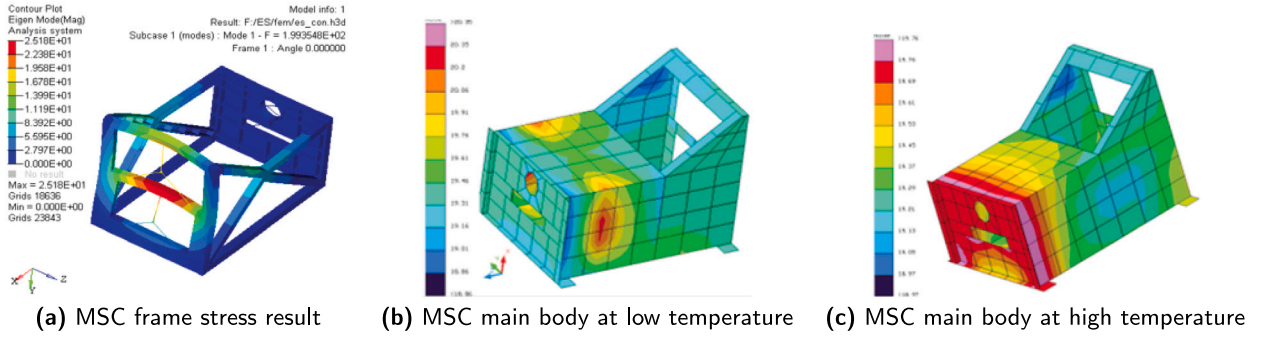


Fig. 9. ETRSS-1 MSC frame stress and main body thermal result.

requiring the scanning mechanism of the optical instrument to perform only a one-dimensional scan in the cross-track direction [5,18].

Depending on the techniques used in receiving and recording the incoming signals, imaging systems are classified into three scanning types: whisk-broom, push-broom, and snapshot [22].

The Ethiopian Earth observation satellite is equipped with a push-broom optical camera with a linear array of detector elements known as a line imager, which scans one line across the sweep width per integration time.

It was chosen because it moves across the track for longer measurements over a specified region and does not require a scanning mirror, resulting in increased reliability and longer life expectancy. In addition, the advantage of push-broom imaging is that a linear CCD can scan a large coverage area and use it to generate a larger image [23], but the FOV requirements, mass, and volume of lenses employed are smaller than those of other imaging systems [24].

Fig. 3 depicts the ETRSS-1 push-broom optical multi-spectral scanner characteristics, which have multiple linearly arranged detector elements that can scan an entire line over a swath at a time.

In addition, earth observation satellites utilize either CCD or CMOS imaging sensors, which have two fundamental functions: converting incoming photons to electrons and converting charge into voltage (a voltage signal). CCDs and CMOS imaging sensors differ in how these functions are performed. The image sensor selection can impact the imager's SNR performance over the spectrum range [25], and it is essential to have high saturation capacity and low noise to ensure that the SNR is sufficient. The Ethiopian earth observation satellite utilized an AT71547(E2V) CCD imaging sensor with a pixel size of $13 \mu\text{m} \times 13 \mu\text{m}$, which converts photons to electrons in pixels and can perform time delay integration without external support. The CCD imaging sensor was chosen because of its high quantum efficiency, dynamic range, intensity, low read noise, and low dark current [24]; however, its limitations include a complex power supply, high power consumption, and poor radiation performance.

3. Results and discussion

3.1. Optical imaging performance analysis

Many-characteristics performance indices can be created to compare optical equipment with similar performance characteristics [18]. For high-resolution spatial optical sensors, three basic variables demonstrate the image's quality (i.e., its corresponding information content).

These variables are the signal-to-noise ratio at spatial frequency zero (i.e. high SNR corresponds to high information content), the modulation transfer function of an instrument at the Nyquist frequency (high MTF corresponds to high information content for sampling rates between zero and Nyquist frequency), and the ground sample distance GSD (i.e. small GSD corresponds to high information content) [5].

The SNR is an inherent limitation in radiometric intensity measurement, whereas MTF represents an optical system's transfer quality

in terms of spatial frequency, allowing the system to respond to the signal depending on its variation speed [26]. As a result, the higher spatial frequencies are filtered through the system, resulting in blurred patterns in the associated image.

3.1.1. Modulation transfer function

The modulation transfer function of an optical imaging system is an important design parameter because it affects the signal modulation induced by each component of the system. It is defined as the magnitude of the Fourier transform of the PSF and the ratio of ground scene intensity variations to image intensity variations at a given spatial frequency, and is more useful than PSF as a design and sensor testing tool for two reasons [9]: First, in a linear system, the individual sensor subsystem MTFs can be multiplied to calculate the net sensor MTF. Second, the modulation transfer function can be calculated simply from measurements of easily designed test targets. In addition, several modulation transfer function components (as illustrated in Fig. 11) must be considered to translate the net desired imaging resolution given by the application requirement to sensor MTF specifications. The optics, the detector FOV, slight electrical and optical crosstalk between detectors, detector temporal response, electronics integration, and frequency response, all contribute to a decrease in image resolution [9]. All of these effects combine to give net sensor resolution performance, which can be reasonably expected based on sensor design and provided design resolution requirements, the associated net modulation transfer function may be budgeted among the various components to get the required result.

$$M_{cm} = M_o \times \kappa \times M_c \times M_e \quad (21)$$

where M_{cm} , is camera MTF, M_o optical MTF, κ camera manufacture and alignment coefficient, M_c CCD/detector MTF, and M_e electronic devices MTF. The net MTF of the entire imaging chain is just the product of the individual MTFs (stated in Eq. (21)) for each component of the chain, which is the product of the detector FOV, the optics, and the electronics modulation transfer function. Optical modulation transfer function is the MTF of the entire lens system, not the result of individual lenses, as these MTFs can be linked together. The diffraction-limited modulation transfer function for circular aperture is given as reported in Ref. [1]:

$$M_d = \begin{cases} \frac{2}{\pi} \left[\cos^{-1} \left(\frac{f}{f_{oc}} \right) - \frac{f}{f_{oc}} \sqrt{1 - \left(\frac{f}{f_{oc}} \right)^2} \right], & \frac{f}{f_{oc}} < 1 \\ 0, & \text{else} \end{cases} \quad (22)$$

where f is spatial frequency, $f_{oc} = \frac{D}{\lambda}$ is optical cutoff frequency, λ radiation wavelength, and D is aperture diameter.

Other important modulation transfer function contributions are time integration smearing and optical blurring (i.e. imperfect geometrical focus), which interact multiplicatively to lower MTF, with the net MTF resembling a Gaussian function as given in Eq. (23).

$$M_{ob} = \exp \left(\frac{-\pi f^2}{\alpha^2} \right) \quad (23)$$

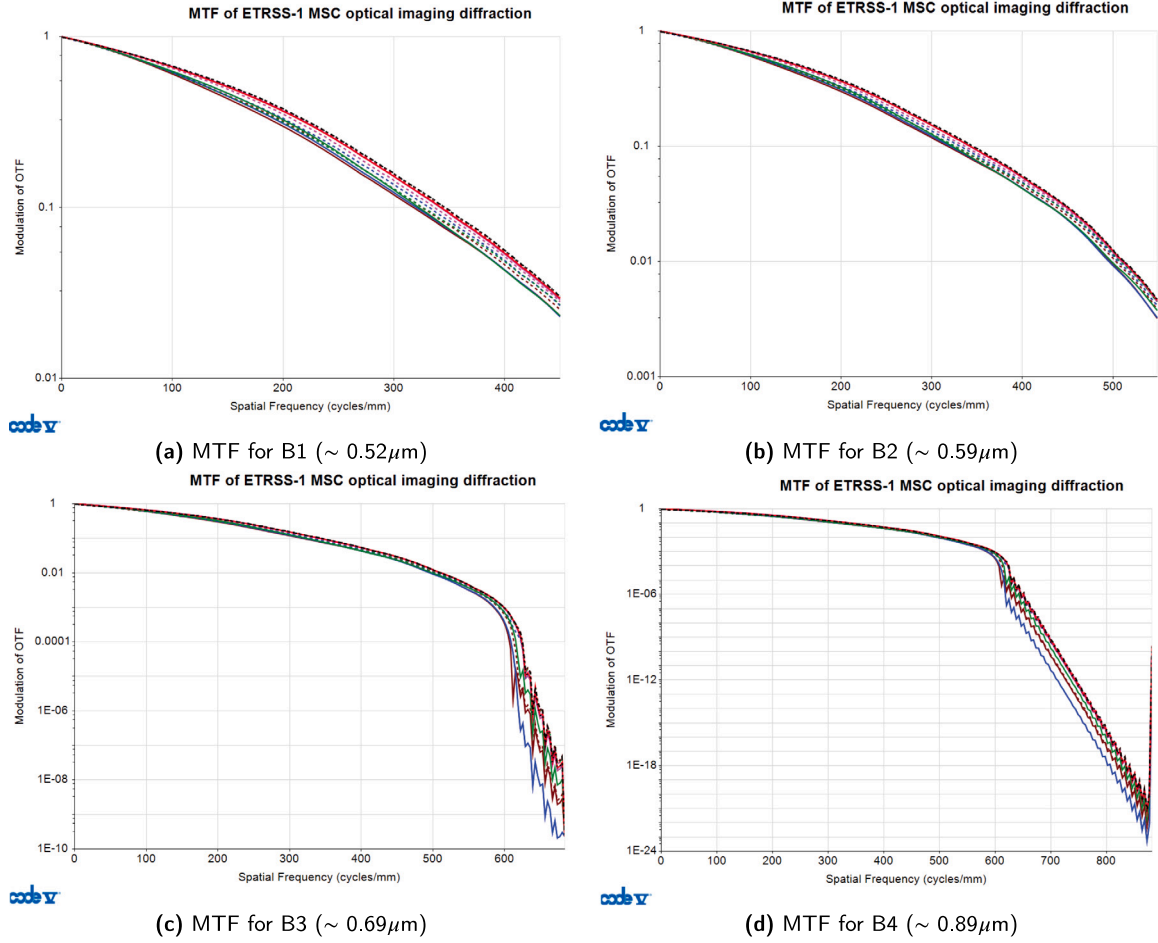


Fig. 10. ETRSS-1 MSC optical imaging payload at center for wavelength 0.45~0.89 μm depending on the spatial frequency (pair lines per mm).

where the parameter α defines the Gaussian function. Thus, the optical imaging system MTF is approximated as given in Eq. (24), assuming a perfect symmetrical system.

$$M_o = M_d \times M_{ob} \cong \left(1 - \frac{f}{f_d}\right) \exp\left(\frac{-\pi f^2}{\alpha^2}\right) \quad (24)$$

whereas Eq. (22) is linearly approximated and can be rewritten as

$$M_d = 1 - 1.27 \left(\frac{f}{f_{oc}}\right) = 1 - \left(\frac{f}{f_D}\right) \quad (25)$$

and the diffraction cutoff (f_D) of the entrance pupil aperture is given $f_D = \frac{f_{oc}}{1.22} = \frac{D}{1.22\lambda}$. It appears that the resolution effect associated with

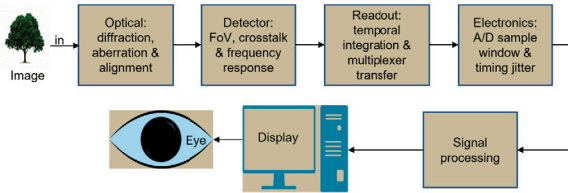


Fig. 11. Optical sensor MTF analysis process.

the primary detection process is determined by the type of system, array format, signal processing, and display type [27], all of which may influence the final system MTF. Furthermore, a rectangular sampling function, such as a uniform detector with a width that defines the sensor FOV, has a sine-function MTF reported in Ref. [9]:

$$M_c = \frac{\sin(\pi \times FOV \times f)}{\pi \times FOV \times f} \triangleq \frac{\text{sinc}\pi f}{\pi f} \quad (26)$$

Table 4
MSC imaging payload MTF parametric value.

MTF index	B_1	B_2	B_3	B_4
Optical	0.749	0.715	0.669	0.590
Mechanical (κ)	0.85	0.85	0.85	0.85
CCD	0.50	0.50	0.50	0.48
Electrical	0.98	0.98	0.98	0.98
Other	0.95	0.95	0.95	0.95
Total static MTF (camera)	0.296	0.282	0.264	0.224

The rectangular detector MTF is unity at zero spatial frequency and reduces to zero at $f = 1/FOV$, with all detector elements assumed to be square or rectangular. If a scanning detector of angular dimension ϕ is temporarily integrated for a time t_i , that integration has an equivalent spatial dimension of $\phi_i = \frac{\phi t_i}{\tau_d}$, where τ_d is the scan dwell time corresponding to angular dimension ϕ . The resulting MTF is $M_s = \text{sinc}(\phi_i f)$. Furthermore, if this or another electrical MTF is substantial in a bidirectionally scanning device, the MTF loss from the corresponding phase transfer function must be accounted for. Thus, the MTF for a single-RC low pass filter and high pass filter respectively are given as reported in Ref. [27].

$$\begin{bmatrix} M_l \\ M_h \end{bmatrix} = \begin{bmatrix} \left(1 + \left[\frac{f}{f_{oc}}\right]^2\right)^{-1/2} \\ \left(\frac{f}{f_{oc}}\right) \left(1 + \left[\frac{f}{f_{oc}}\right]^2\right)^{-1/2} \end{bmatrix} \quad (27)$$

However, a charge transfer device, such as one utilized in an optical imaging system's focal plane readout, a time delay and integrated (TDI)

circuit, or an analog memory, has an MTF due to nonunity charge transfer efficiency, as shown in Eq. (28).

$$M_t = \exp \left(-m - (1 - \epsilon)(1 - \cos(2\pi \frac{f}{f_s})) \right) \quad (28)$$

where m is the number of gates in the transfer from the detector to the output amplifier, ϵ is the charge transfer efficiency of each gate, and f_s is the sampling frequency of the structure. To summarize, the total system MTF of an optical imaging system is the sum of the subsystem functions outlined above.

Fig. 10 shows the logarithmic diffraction MTF modulation for four bands in the Ethiopian Earth observation satellite MSC optical imaging payload. Further, the Ethiopian Earth observation satellite MSC optical imaging payload has a reasonable net MTF of more than 20%, with individual MTF for each component described in Table 4. The average MTF for each spectral band of the Ethiopian Earth observation satellite MSC optical imaging payload is for B1 0.7351 μm , B2 0.7302 μm , B3 0.6740 μm , and the minimum MTF of the optical system is 0.587 μm for B4.

3.1.2 Signal to noise ratio

The signal-to-noise ratio (SNR) and modulation transfer function (MTF) are the two most important characteristics impacting the image quality of earth observation satellite sensors. The signal-to-noise ratio is a statistical metric that describes the optical sensor's strength to distinguish signal from noise [9]; higher numbers mean greater performance and an ideal sensor would have infinite SNR. Several noises contribute

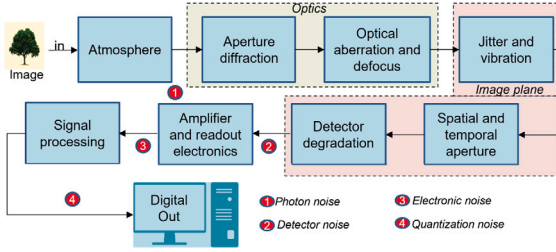


Fig. 12. Optical sensor SNR analysis process.

to a decrease in an optical imaging system's signal-to-noise ratio [7,9], including photon noise, detector noise, readout electronic noise, and digital quantizer, all of which contribute to the degradation of desired image quality.

Fig. 12 depicts a schematic explanation of optical elements from the target image to the output, as well as noise components, where the process must account for image fidelity loss as a function of spatial frequency or spatial resolution. The signal detection analyzes the optical sensor's detection performance in the presence of noise, whereas imaging quality quantifies the efficiency of transmitting spatial information from the target via electronic image capture [1]. The signal-to-noise ratio (SNR) is calculated by dividing the average signal level by the noise standard deviation (σ) at that level represented in Eq. (29).

$$SNR = 20 \log \left(\frac{S_e}{\sigma} \right) \quad (29)$$

where S_e and σ are the numbers of the output signal and the electron number of the noise signal. The noise standard deviation (σ) at any signal level is a combination of signal-dependent and signal-independent noise sources. It is determined by summing the variances of the four noise sources described in Eq. (30).

$$\sigma = \sqrt{\sigma_p^2 + \sigma_d^2 + \sigma_e^2 + \sigma_q^2} \quad (30)$$

where σ_p is the circuit (photon) noise, σ_d detector noise, σ_e electronics noise, and σ_q quantization noise.

Table 5
MSC imaging payload SNR analysis.

Bands	Sun elevation: 20°, Surface albedo ρ : 0.05		Sun elevation: 70°, Surface albedo ρ : 0.65	
	Radiance of entrance pupil/W m ² sr ⁻¹	SNR	Radiance of entrance pupil/W m ² sr ⁻¹	SNR
B1	2.3585	34.6 dB	25.0570	49.4 dB
B2	1.5932	32.9 dB	22.5617	49.7 dB
B3	0.8966	29.4 dB	16.3257	48.7 dB
B4	0.9678	28.2 dB	22.2701	49.1 dB

Photon noise: The number of electrons generated in the detector in response to an incident light radiant power signal from the optics is proportional to the incident light signal, which subsequently becomes the electron signal that leaves the detector [9]. From a statistical standpoint, the light follows a poisson distribution, which means that the noise variance is equal to the signal and the standard deviation is quantitatively equal to the square root of the signal (i.e. $\sigma^2 = S_e$). Thus, the output signal (S_e) is estimated in electrons per sample period, where the sample period is T_{int} and the detector is allowed to integrate light radiance reported in Ref. [16]:

$$S_e = \frac{\pi A_d T_{int}}{4(F_l/D)^2 hc} \int_{\lambda_{min}}^{\lambda_{max}} \eta(\lambda) L(\lambda) \tau_o(\lambda) \lambda d\lambda \quad (31)$$

where A_d is the area of the detector pixel [cm²], T_{int} integration time [s], f focal length [cm], D optical aperture diameter [cm], $L(\lambda)$ spectral radiance at optical entrance pupil $\left[\frac{\text{W}}{\text{cm}^2 \mu\text{m sr}} \right]$, $\eta(\lambda)$ quantum efficiency of the detector in electrons per photon, $\tau_o(\lambda)$ transmittance of the optical system, h Planck constant, and c velocity of light in the vacuum. Jafarsalehi et al. [7] provides a formula for calculating the total spectral radiation at the optical entrance pupil and orbital altitude, R .

$$L(\lambda) = L_p + \frac{\rho_\lambda \times \tau(\lambda) \times E_\lambda}{\pi} \quad (32)$$

where E_λ is the solar/lunar spectral radiance at orbital altitude from the Earth's surface or target, $\tau(\lambda)$ is called atmospheric transmittance in the path (target to sensor), ρ_λ is the spectral reflection coefficient of target, and L_p is the sum of the multiple scattered solar and atmospheric emitted from radiance into the path. Table 5 presents the signal-to-noise ratio of the ETRSS-1 MSC optical imaging payload's spectral bands under various sun elevation circumstances.

Detector noise: The kinetic energy of electrons in a material with resistance R is partly provided by thermal energy, which is directly proportional to absolute temperature T , with the proportionality constant given by Boltzmann's constant k . Subsequently, the detector noise is determined:

$$\sigma_d = \frac{\sqrt{2kT T_{int}/R}}{q} \quad (33)$$

where q is the charge on an electron, equal to 1.6×10^{-19} C.

Electronic noise: The noise in electrons/samples from the analog detector readout circuits depends on the specific circuitry used in the sensor. There is no simple expression to describe the noise performance of analog circuits. However, experience with such circuits shows that a nominal value from $\sigma_e = 50 \sim 200$ electrons/sample [9] is typical with special electronics designs capable of as low as a few electrons per sample.

Quantization noise: Regardless of the detector signal level or the noise level of the detector and analog electronics, the analog-to-digital (A/D) converter contributes a signal-independent noise component due to uncertainty in the least significance bit (LSB) of quantization. Subsequently, the quantization noise estimate is reported in Refs. [7,9]:

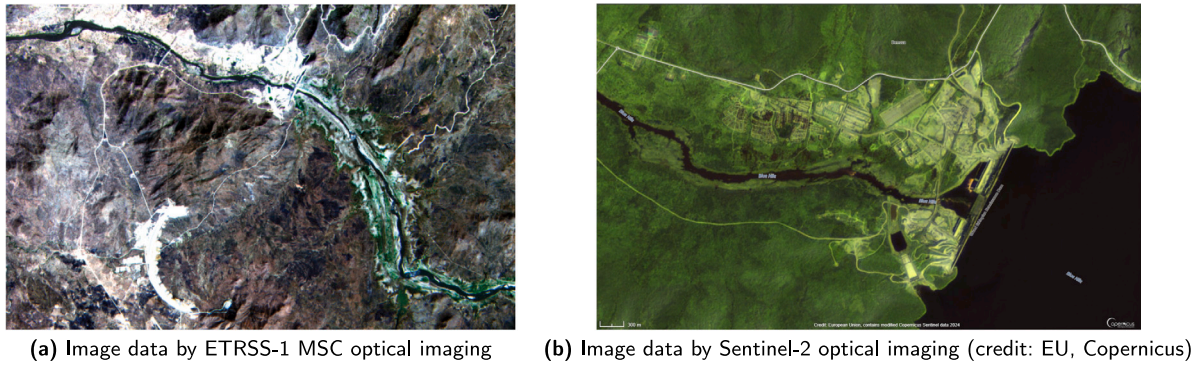


Fig. 13. Imagery data acquired from GERD by ETRSS-1 MSC optical imaging camera in contrast to Sentinel-2.

$$\sigma_q = \frac{N_{max}}{k_s \sqrt{122^{b-\epsilon}}} \quad (34)$$

where N_{max} is the saturation electron signal corresponding to the maximum expected light radiance input, b number of bit of quantization, ϵ encoder noise bit degradation (typically 0.5), and k_s factor to avoid A/D converter saturation (typically 0.9)

3.2. On-orbit performance analysis

3.2.1. Data collection and compression efficiency

The efficiency with which the Ethiopian earth observation satellite's multispectral imaging optical camera collects and compresses data is critical for making the most effective use of limited onboard storage and downlink bandwidth resources. The data transmission subsystem of the ETRSS-1 has the function of recording the MSC optical imaging camera data via a storage device such as solid state memory, compressing the data from the MSC, and transmitting it to the ground station [3] directly via an X-band phased array antenna. The ETRSS-1 spacecraft's MSC optical imaging data must be compressed before transmission to fit within the available data rates of the data transmission subsystem. The ETRSS-1 spacecraft DTS has a channel data rate of 100 Mbps, SSM's writing rate is 140 Mbps, and the raw data rate from the MSC optical image camera and auxiliary is 133.6 Mbps. However, the rate after compressing is $133.6 \text{ Mbps} / 1.25$ (lossless compression) = 106.88 Mbps. The higher the compress ratio, the more images may be transmitted to the ground station at the same time, however the lower the compress ratio, the higher the restored image quality.

Fig. 13 illustrates the optical imagery taken by the Ethiopian Earth observation satellite MSC optical imaging camera over the GERD in comparison to the images collected by Sentinel-2 optical sensors. It is crucial to emphasize that the comparison is not an in-depth evaluation of the two spacecraft's outputs, but rather a preliminary analysis indicating if the images collected are comparable or visually deviate

from one other. Additionally, it is worth noting that the spacecraft's two optical imaging payloads differ in terms of design, the altitude at which imagery data is collected, spatial resolutions (e.g., 10 m for Sentinel-2 and 13.75 m for ETRSS-1), and other orbital parameters. The results show that the imagery data collected by the Ethiopian Earth observation satellite is comparable to Sentinel-2 images taken over the same locations (i.e. GERD, located in Ethiopia's Benishangul-Gumuz Region, roughly 14 km east of the Sudanese border).

3.2.2. MSC operational mode and application

The Ethiopian Earth observation satellite's MSC imaging payload typically operates in three modes during launch and in orbit: real-time transmission, recording, and playback. Furthermore, during the launch mode, the MSC optical imaging payload of the ETRSS-1 is powered off and there is no thermal control; however, there is a standby mode in which the MSC camera is powered off [2], does not take images, and the OBHD subsystem of the spacecraft maintains the MSC imaging payload thermal control; as well as in the imaging mode, the MSC is powered on and acquires an image of the target in the nadir with a roll angle $\leq \pm 35^\circ$ and $\pm 90^\circ$ yaw angle. When the satellite is within the ground station's reception range, the ground measurement and control system sends a non-side sway real-time transmission command block to the satellite and receives image data in real time. The spectral application fields of the Ethiopian Earth observation satellite MSC optical imaging payload have been explained in Table 6.

3.2.3. Radiation and geometric correction accuracy

There are two types of calibration that are important for optical and other types of spacecraft imaging sensors: intensive prelaunch calibration and characterization to provide users with the best connection between the sensor radiation measurements and the FOV, and periodic calibration after launch into orbit given that the sensors degrade over time and temperature. All calibration efforts aim to establish and maintain over time the relationship between satellite radiance and the in situ characteristic being measured. On-orbit calibration updates are

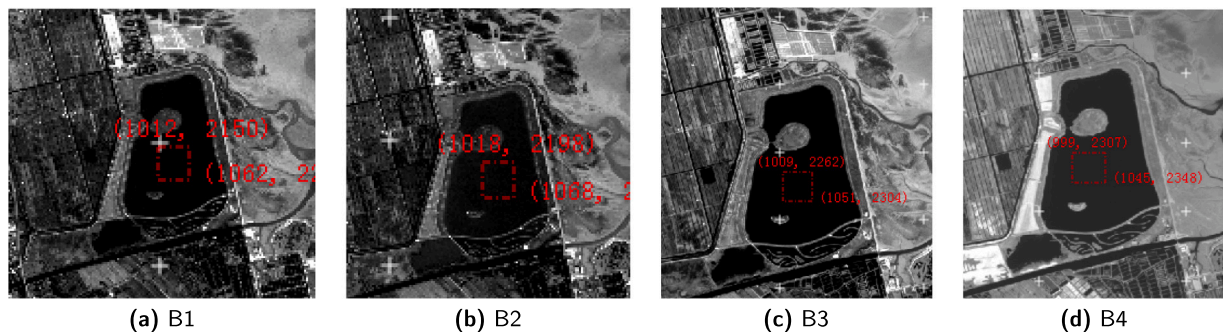


Fig. 14. Low area selection of ETRSS-1 MSC for band-1 to band-4.

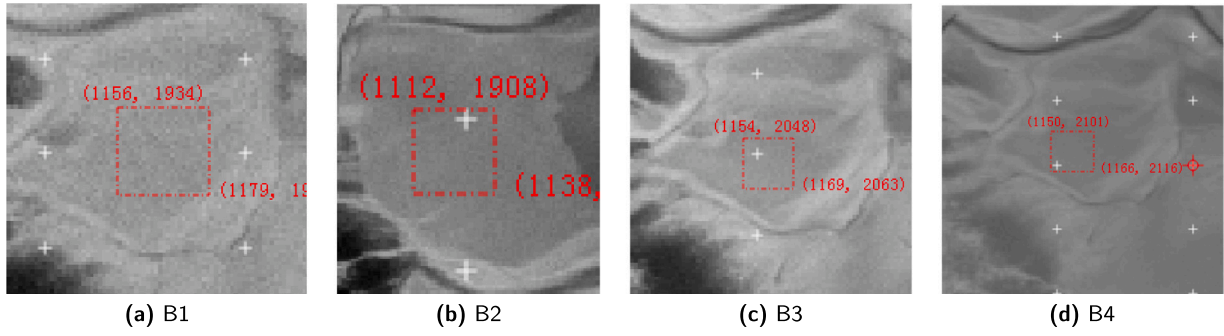


Fig. 15. Medium area selection of ETRSS-1 MSC for band-1 to band-4.

Table 6

Main application fields of ETRSS-1 MSC imaging payload.

Spectrum	Range [μm]	Applications
B1	0.45~0.52	Water body and coastal mapping, soil/vegetation differentiation, forest type mapping, and feature identification.
B2	0.52~0.59	Evaluating green reflectance of vegetation and identifying features.
B3	0.63~0.69	Vegetation analysis and differentiation of plate types
B4	0.77~0.89	Determining vegetation types vigor and biomass survey delineating water bodies and soil moisture differentiation

needed to account for response changes in orbiting instruments due to aging, contamination, and surface deterioration [28]. As a result, streaking noise occurs on the image, generating target deformation and other visual effects that influence target perceptions. Furthermore, the image should be corrected to enhance its visual quality and identify the target. This process, known as radiometric processing, is required to obtain high-quality optical satellite imagery from sensors with several detectors per band [29]. The relative radiation calibration accuracy is given:

$$\sigma_r = \frac{\delta}{S} \quad (35)$$

where δ is the standard deviation of the corrected image gray value, and S is the mean value of the corrected image gray value. However, the mean value represents the average feature of the image population and is calculated as:

$$\bar{I} = \frac{\sum_{i=1}^M \sum_{j=1}^N I(i,j)}{M \times N} \quad (36)$$

where $I(x,y)$ is the image function, and M and N are the numbers of rows and columns of the image respectively. The standard deviation is also known as the gray value variance algorithm, and it is calculated

Table 7

Summary of relative radiation correction accuracy test data results of ETRSS-1 MSC imaging payload.

Band name	Low reflection area [%]	Medium reflection area [%]	High reflection area [%]
Band-1	0.8902	0.8219	0.7608
Band-2	1.0334	0.9179	0.8013
Band-3	1.4771	1.3772	0.7603
Band-4	1.8362	1.7048	1.2337
Mean value	1.3092	1.2054	0.8890
ARC ^a			1.1346

^a ARC: The accuracy of relative radiation correction.

as stated in Eq. (37). The algorithm uses the gray level average of all the pixels in the image as a reference, calculates the squared sum of each pixel's gray value, and normalizes the total number of pixels. It shows the level of grayscale change in the image. The higher the average degree of grayscale change, the clearer the image; the smaller the average degree of grayscale change, the more blurred the image.

$$SD = \sqrt{\frac{\sum_{i=1}^M \sum_{j=1}^N (I(i,j) - \bar{I})^2}{M \times N}} \quad (37)$$

Figs. 14, 15, and 16 shows the relative radiation correction accuracy of the results of test data of the MSC optical imaging payload of the Ethiopian earth observation satellite. Furthermore, the test data selection includes four MSC spectrum bands, as well as an imaging area spanning different latitudes and longitudes. The selected areas are low reflection area, medium reflection area, and high reflection area, with at least 100 pixels in each area. Table 7 summarizes the relative radiation correction accuracy test data results from the MSC optical imaging payload of the Ethiopian Earth observation satellite. It is important to note that, from the analysis of the test data acquired by the MSC of the Ethiopian Earth observation satellite, the relative radiation accuracy is less than 3% which satisfies the requirements. Another key component that can help improve image quality is geometric calibration with references such as ground control points (GCPs), coastlines, and stars.

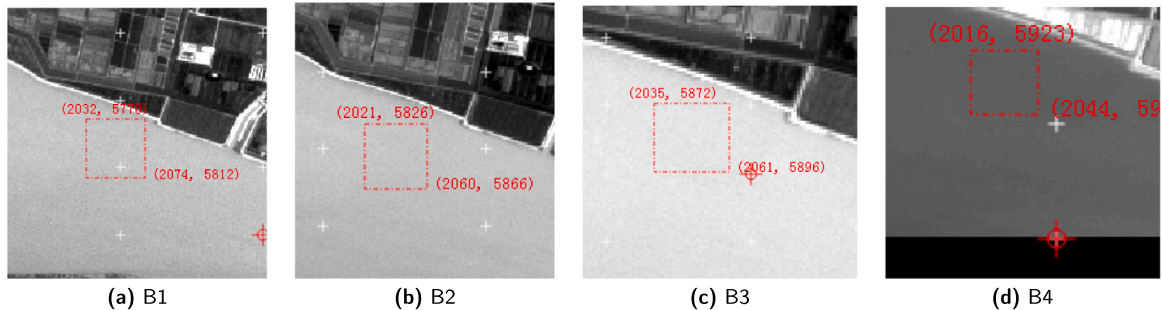


Fig. 16. High area selection of ETRSS-1 MSC for band-1 to band-4.

This is a necessary prerequisite for providing high-precision satellite images [30,31]. The controlled geometric positioning accuracy of each camera image of the Ethiopian earth observation satellite is evaluated using the circular probability error (CE90) method, which selects ten feature points for a scene. The statistical analysis of the test data results is based on the root-mean-square that is generated from the error of the ten test locations, and the Ethiopian Earth observation satellite MSC optical imaging camera single loaded controlled positioning accuracy was 22.82124 m, which is computed using

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\Delta d_i)^2} \quad (38)$$

where Δd_i is the error at each point and N is 10. It is essential to note that when 9 feature points achieve an accuracy index of 3 pixels CE90, the scene image's controlled geometric positioning accuracy meets the index's requirements. Furthermore, the ETRSS-1 spacecraft MSC optical imaging camera image product has a control geometric positioning accuracy test method: ENVI and ArcGIS software display the product image and the reference image to be tested, and similar features between the image to be tested and the reference image are manually identified.

To summarize, based on verification of the selected test data, the controlled geometric positioning accuracy of Level4 orthorectification image products of ETRSS-1 MSC optical imaging camera satisfies the accuracy requirement of 3 pixels CE90 (that is, more than 90% of each product is better than 3 pixels).

4. Conclusion

The Ethiopian Earth observation satellite's MSC optical imaging payload, comprising push broom scanners, has been deployed in a sun-synchronous polar orbit at an altitude of 628.10 km. Nonetheless, it is critical to understand the underlying design aspects when designing an optical imaging system that meets the specific requirements. The top-down systematic approach has been utilized for investigating the design of the Ethiopian earth observation satellite MSC optical imaging payload in terms of system performance requirements, such as SNR (detection) and MTF (resolution). Additionally, Ethiopia's earth observation satellite optical MSC imaging type design, performance characteristics, optical, mechanical, thermal, data transfer, interface, radiometric and geometric calibration, and image quality have been presented.

The optomechanical design and thermal simulation results for a satellite optical imaging payload with 13.5 m spatial resolution and four bands are presented as well, and the results demonstrate that the requirements are satisfied. Furthermore, an optical system based on unobscured off-axis TMA is then successfully designed, having an off-axis TMA system with a FOV of $7.4^\circ \times 0.3^\circ$, a relative F-number of 9.8, and a spectral range of 0.45–0.9 μm . The results show that the total average MSC optical imaging payload camera SNR at sun elevation 20° and surface albedo of 5% is 31.27 dB, while at sun elevation 70° and surface albedo of 65% is 49.2 dB, satisfying the mission requirements. The Ethiopian Earth observation satellite MSC's optical imaging payload MTF is greater than 20%, meeting the design requirement. However, the optics, detector FOV, slight electrical and optical crosstalk between detectors, electronics integration, and frequency response all contribute to a decrease in image resolution and modulation transfer function. Furthermore, an optical payload system image quality of the Ethiopian Earth observation satellite is within the diffraction limit, with a maximum relative distortion of 13.26%. This indicates the optical imaging system performs well in practical applications.

The on-board performance analysis results in terms of radiometric and geometric calibration show that the payload's imagery is georeferenced and aligned with geographical coordinates to provide high-quality imagery for a variety of applications, including environmental monitoring, agriculture, urban planning, and disaster management.

CRediT authorship contribution statement

Dinaol Gadisa: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Eshet Tafes:** Writing – review & editing, Visualization, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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